ASHA Vaani

A voice field register for ASHA workers

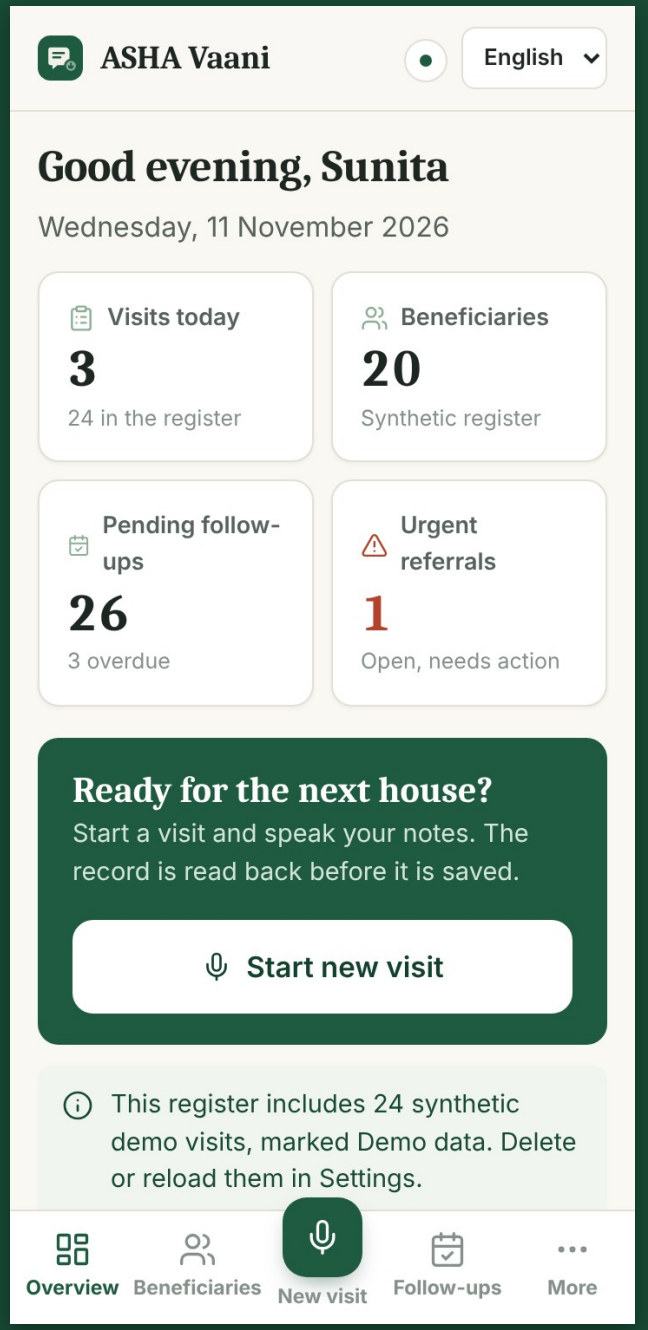
Today, every home visit ends with typing

- Pregnancy, newborn and immunisation visits are logged in paper registers and phone apps, on small screens, while walking between houses.
- Data entry eats into visit time, and records arrive late or incomplete.
- A missed danger sign, such as bleeding in pregnancy or a newborn with high fever, can cost a life.

Low-end Android phone

Household and street noise

Hindi or Kannada + English health words



Speak once, confirm by voice, refer on time

Speak

Dictate the visit naturally, code-mix welcome

Hear it back

Form read aloud, numbers spelled out

Fix by voice

“No, weight is 54 kilo”: only that field changes

Confirm

Saved only after an explicit yes

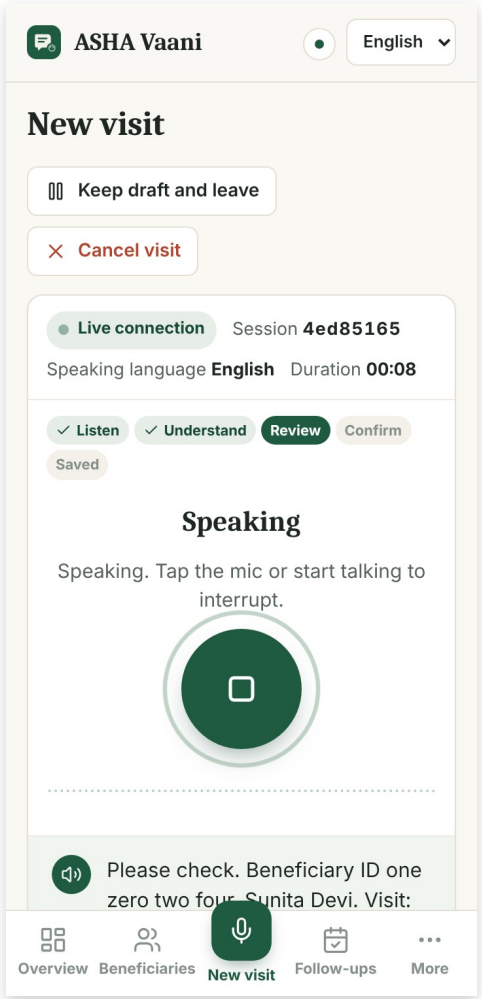
Next actions

Follow-ups; urgent referral first

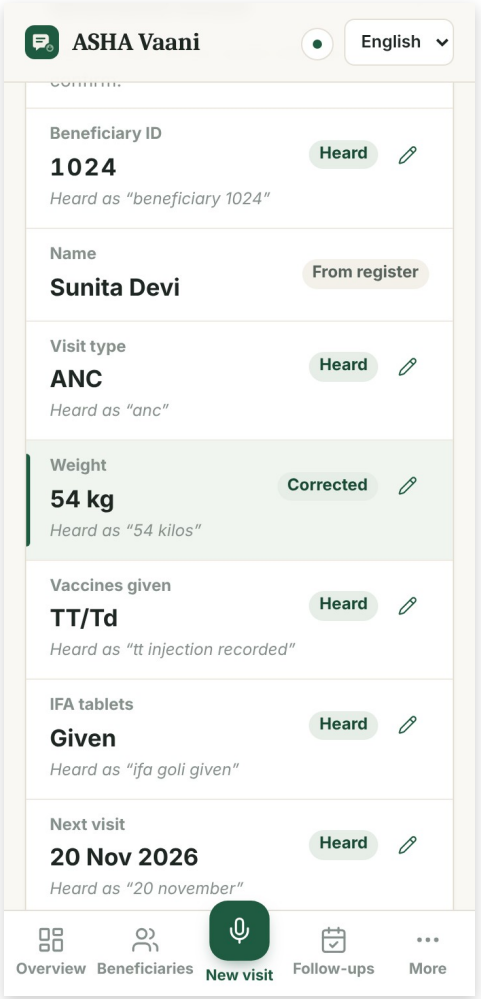
Danger signs interrupt the flow

When the notes mention a sign from the MCP Card 2018 list, the agent immediately says, in her language, to refer to the PHC now, and flags the record. It never diagnoses or suggests medicine. Negated mentions (“bleeding nahi hai”) are not flagged; unclear ones get one yes/no question.

One visit, start to finish, without typing



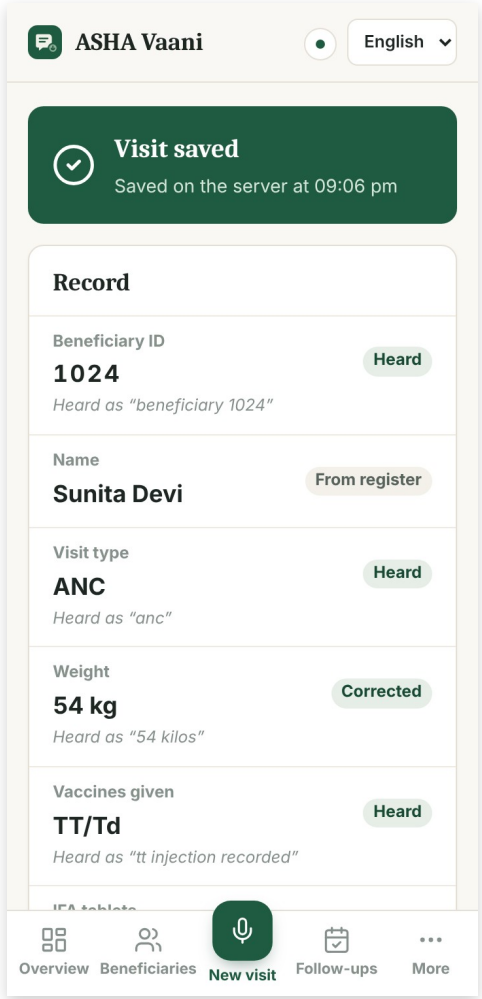
Read-back of what was heard



Voice correction: weight only

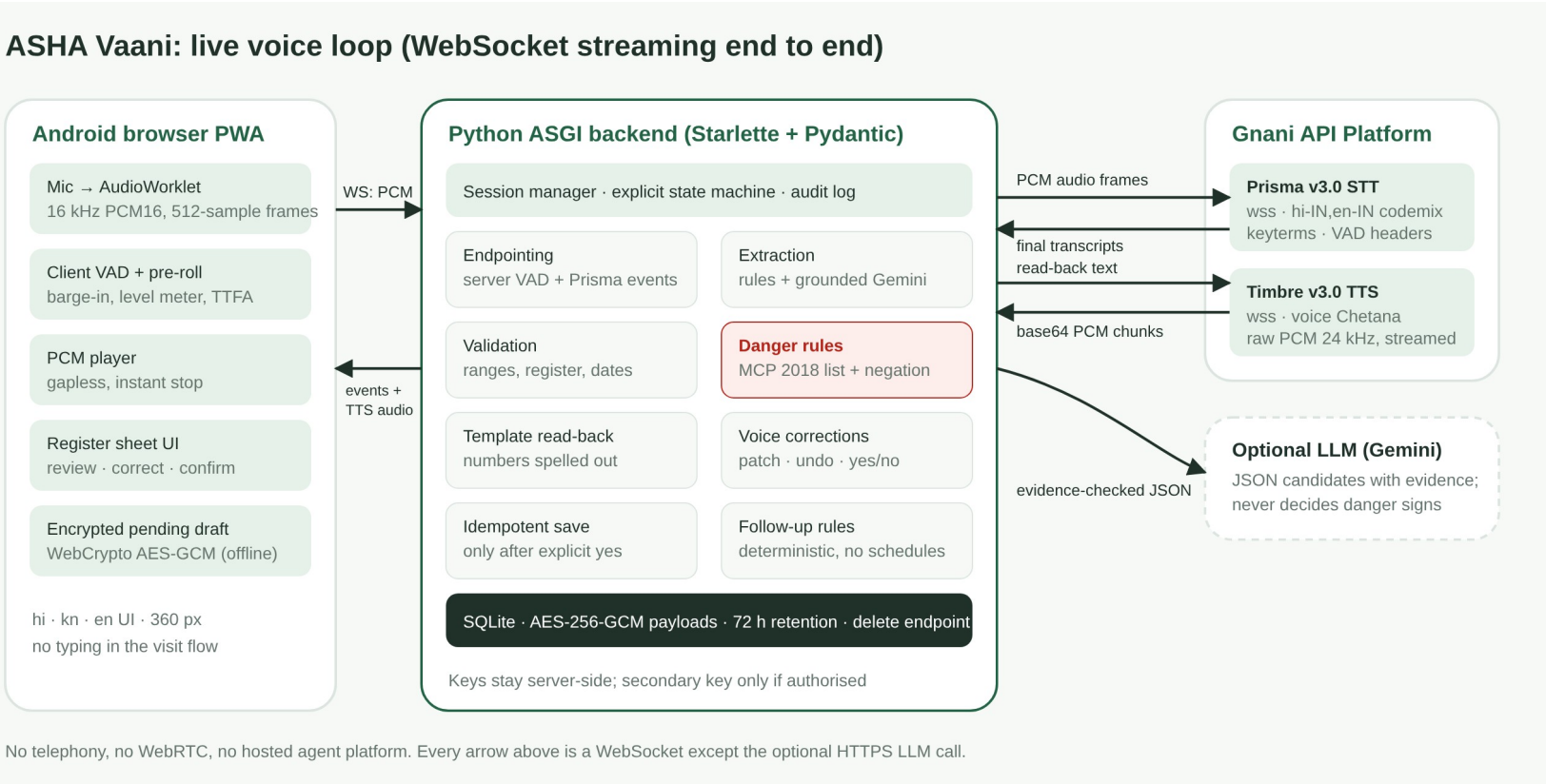


Danger sign in Hindi: refer now



Saved, with next actions

Streaming end to end on Gnani Prisma v3.0 and Timbre v3.0



Speech-layer depth

- 1,024-byte PCM frames at real-time cadence; final transcripts only, never invented partials
- Endpointing tuned so pauses inside spoken IDs are not cut off
- Barge-in: playback stops, Timbre stream dropped, 320 ms pre-roll kept
- Code-mix: hi-IN,en-IN / kn-IN,en-IN plus medical keyterms; Hindi and Kannada number words
- Every stage timed: median and p95 in the app

Custom orchestration (no hosted agent platform, no telephony). Optional Gemini extraction must quote evidence found in the transcript and can never raise or clear a danger flag.

Safe by design, measured honestly

16/16
danger signs caught

0
false danger alerts

0
invented fields

10/10
voice corrections applied

On our 65-utterance scripted set (Hindi, Kannada, English, code-mix, noisy text), written alongside the rules. It shows the rules work as designed; live accuracy and latency are measured in-app on real speech. Plus 175 backend and 6 frontend automated tests.

Deterministic safety

- Fixed MCP Card 2018 list (MoHFW and MoWCD)
- Negation, “not stopping”, uncertainty, past mentions
- Alert spoken before the read-back

Responsible AI

- Says it is an AI in its first turn
- Consent before recording; synthetic data only
- No diagnosis, medicine or dosage advice

Honest by construction

- “Saved” only after the database commits
- AES-256-GCM at rest; deleted after judging
- Offline drafts labelled “pending, not saved”

A correct record before the next house

Try it live

1. Dictate a visit in any of our languages, with noise
2. Correct one value by voice, or say undo
3. Say “haan” and watch the saved record and next actions
4. Mention a danger sign, then a negated one

What comes next

- Health-professional and native-speaker review of the danger phrases and referral wording
- Consented field pilot to measure real accuracy and time saved per visit
- More of Prisma’s 10 Indian languages, once each path is tested
- Worker login, ANM review queue and export to official registers

Built on Gnani Prisma v3.0 and Timbre v3.0 · synthetic data only · not clinically validated